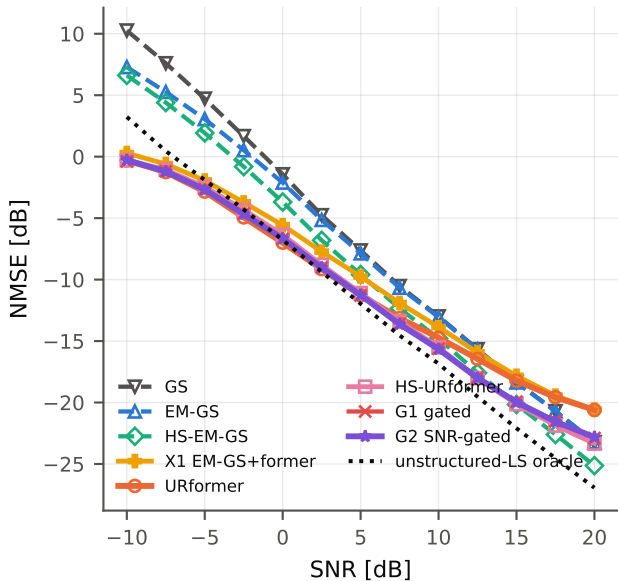
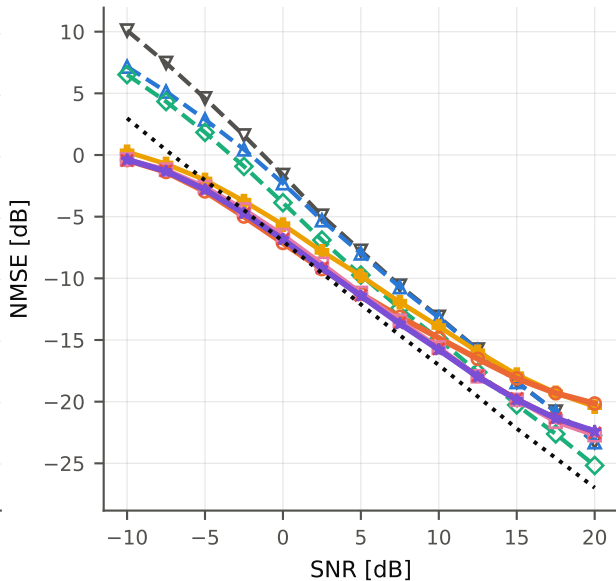


NMSE vs SNR — all methods (P = 20 pilots)

a. per-trial MEDIAN (project primary)



b. ratio-of-sums (paper convention)



K=3 users, M=32 atomic antennas, RSR=10 dB (repo convention; 5.23 dB in the paper's multi-user convention), $L_k \sim \mathcal{U}\{3..7\}$ paths, 400 trials per point. Classical arms dashed, learned arms solid. All learned arms trained at P=20 on 80k samples (URformer/HS-URformer/G1/G2) or on cached EM-GS (X1).